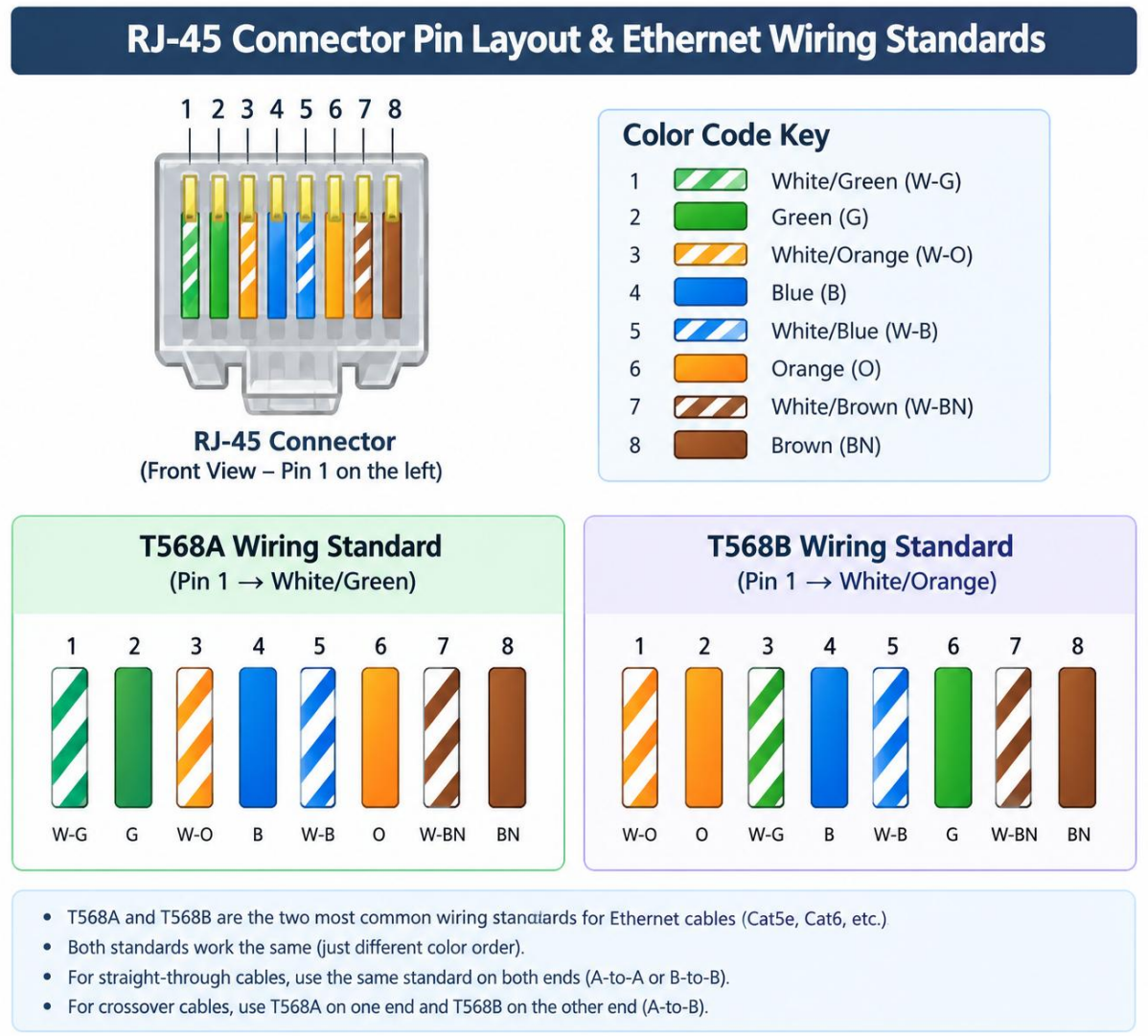
Ethernet standard and specification

Q-1 Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

Ans:



Q-2 Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Ans: Comparison of Ethernet Standards

Specification	**10BASE-T**	**100BASE-TX**	**1000BASE-T**
Speed	10 Mbps	100 Mbps	1 Gbps (1000 Mbps)
Cable type	Cat3 or higher UTP	Cat5 or higher UTP	Cat5e or higher UTP
Maximum cable length	100 metres	100 metres	100 metres

Connector type	RJ-45	RJ-45	RJ-45
Pairs of wires used	2 pairs	2 pairs	4 pairs
Common name	Standard Ethernet	Fast Ethernet	Gigabit Ethernet

Sources: Cisco and Ethernet technical references.

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Q-3 Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type.

Ans: Connecting Two Laptops Directly Using Ethernet

Cable used: Ethernet Crossover Cable

Connector type: RJ-45

Explanation:

A crossover cable is traditionally used to connect two similar devices, such as two laptops, directly without a network switch.

- In a crossover cable, the transmit (TX) and receive (RX) wire pairs are crossed at one end.
- This allows one laptop's transmit signals to reach the other laptop's receive pins, and vice versa.
- A straight-through cable uses the same wiring order at both ends and is traditionally used to connect different types of devices, such as a laptop to a switch or router.

Q-4 List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Ans: Three Devices That Use Ethernet Connections

No.	**Device**	**Likely Ethernet Standard**	**Use**
1	Wi-Fi Router	1000BASE-T	Connects to a computer or network switch for high-speed internet access.
2	Desktop Computer in College Lab	1000BASE-T	Provides a fast and stable wired network connection.
3	Network Printer in College	100BASE-TX or 1000BASE-T	Connects to the college network so multiple users can print documents.